

# Demonstration of a Quantum-Safe Network Service Including Quantum Key Distribution, Software-Define Networking and Telco Edge Cloud Integration

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**Abstract** *This demonstration presents a quantum-safe network service integrating QKD, SDN, and Telco Edge Cloud technologies, enabling secure key distribution, real-time monitoring, and industrial application protection, demonstrating scalability, interoperability, and deployment feasibility in realistic telecom environments. ©2026 The Author(s)*

## Review

Quantum Key Distribution (QKD) enables the secure exchange of cryptographic keys based on fundamental quantum properties such as the no-cloning theorem and measurement-induced disturbance [1-3]. These properties ensure that any eavesdropping attempt introduces detectable errors, quantified through the Quantum Bit Error Rate (QBER) [4]. The secure keys obtained after post-processing procedures [5] like sifting, error correction and privacy amplification can be used in symmetric encryption schemes such as the Advanced Encryption Standard (AES) [6]. However, practical QKD systems are constrained by physical limitations including optical losses in fibre links, detector inefficiencies, and the impossibility of amplifying quantum signals. As a result, QKD links require careful monitoring of key parameters such as Secret Key Rate (SKR) and QBER to guarantee secure operation [4].

From a networking perspective, these constraints highlight the need for advanced control and orchestration mechanisms. QKD networks consist of multiple interconnected nodes and links that must be dynamically managed and monitored to be able to reroute the key forwarding in a meshed network if a QKD link fails longer than the key storage can compensate. Traditional network architectures are not designed to handle such requirements. Software Defined Networking (SDN), such as TeraFlowSDN employed in this demo [7], introduces a programmable and centralized control paradigm, separating the control plane from the data plane [8,9]. This enables global visibility of network resources, dynamic configuration, and policy enforcement.

In parallel, Telco Edge Cloud (TEC) infrastructures enable distributed processing capabilities closer to the data sources, reducing latency to less than 5 ms and improving efficiency in industrial applications [10]. The integration of QKD with

SDN and TEC represents a critical step toward scalable quantum-safe networks.

## Novelty

The main novelty of the proposed demonstration lies in the integration of QKD, SDN, and TEC technologies to enable a quantum-safe communication service characterized by secure key distribution, low latency, and advanced monitoring and control capabilities. This combination enables an end-to-end secure service where quantum-generated keys protect application traffic, SDN ensures centralized control and dynamic network management, and Telco Edge Cloud infrastructures provide proximity-based processing and reduced latency. A dedicated control and monitoring framework provides real-time visibility of key performance indicators, allowing continuous supervision and efficient management of the quantum-safe service.

The demonstration developed by FiberCop within the Innovation Hangar in Turin addresses the growing need for secure, distributed, and low-latency digital infrastructures, particularly for industrial and mission-critical applications. In this context, TEC enables computing and processing capabilities to be placed close to data sources, while QKD introduces a novel approach to cryptographic key generation and distribution based on quantum mechanical principles.

The reference use case focuses on the secure transmission and processing of a video stream originating from an industrial environment. This scenario reproduces a deployment in which data generated at a customer site are securely transmitted to a FiberCop TEC node for local processing by advanced digital applications.

## Relevance for ECOC Audience and Community

The ECOC community focuses on optical

communication systems and advanced networking technologies. In this context, the proposed demonstration is highly relevant as it bridges the gap between independent QKD tests and deployable network solutions in realistic telecom environments.

Unlike traditional approaches that consider QKD as a standalone technology, this work demonstrates its effective integration with SDN and TEC infrastructures, enabling a fully operational quantum-safe service, highlighting how secure key generation, centralized control, and low-latency data processing can coexist within a unified architecture, addressing key challenges related to scalability, interoperability, and operational management. In particular, the introduction of a control framework based on ETSI-compliant interfaces allows centralized supervision of QKD nodes and links, offering visibility on critical metrics such as QBER and SKR and enabling dynamic adaptation of the network behavior.

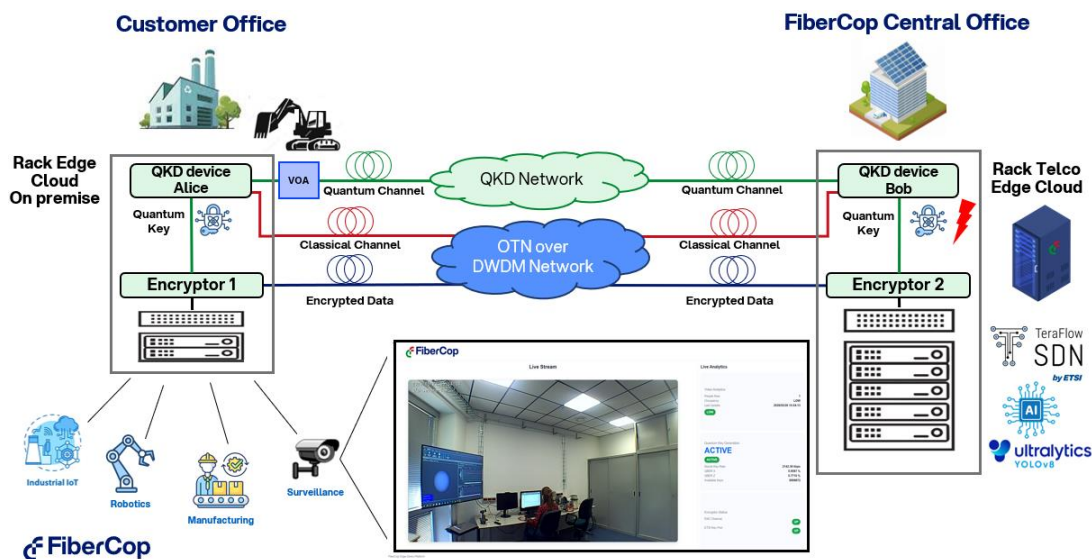
Furthermore, the proposed architecture aligns with ongoing trends in network softwarization, disaggregation, and cloud-native deployments. The industrial video analytics use case provides a tangible example of this integration, showing how encrypted data streams can be securely processed by Artificial Intelligence (AI) applications at the network edge. More broadly, the demonstration illustrates the evolution of QKD from isolated point-to-point links to fully integrated network services, capable of supporting real-world applications and operational requirements. This transformation is particularly relevant for the ECOC community, as it reflects the convergence of optical technologies, programmable networks,

and cloud-based infrastructures. As such, the work provides valuable insights for both academic researchers and industry practitioners interested in the deployment of scalable, interoperable, and quantum-safe communication services.

### Demo Implementation

The demonstration aims to validate an end-to-end quantum-safe communication service integrating QKD, SDN control, and Telco Edge Cloud processing. It showcases video surveillance as an exemplary secure service, where application workloads run on the FiberCop TEC node while connectivity is protected using QKD-generated keys. The service also includes a TeraFlowSDN-based control system that manages and monitors the QKD network as a privileged end-to-end secure connection.

The demo includes the employment of two computers remotely connected to FiberCop Innovation Hangar labs where two QKD Quky devices from ThinkQuantum, Adtran FSP3000 9TCE encryptors, the surveillance camera and the TEC node as well as other supportive instruments are located, simulating the Customer and FiberCop Central Offices respectively (**Fig. 1**). The demonstrative setup is organized across two distinct domains: a quantum channel dedicated to key generation and a classical Optical Transport Network (OTN) over Dense Wavelength Division Multiplexing (DWDM) network carrying encrypted application traffic as well as the classical communication between QKD devices. The latter are located at the two endpoints and they generate and synchronize cryptographic keys, which are then used in combination with classical key alignment



**Fig. 1:** Schematic representation of the demonstration components: the Customer Office where the surveillance camera is located, is connected to the Telco Edge Cloud node employing a classical OTN over DWDM network where the encrypted data are sent, and a quantum channel where the QKD keys are generated. In the dashboard is shown the video stream and the metrics collected by the system. Two events can be triggered by the attendees: the quantum channel interruption through the VOA (represented by the bulldozer) and the keys leakage (represented by the red thunderbolt).

to form hybrid keys by employing hardware-based AES encryption engines in Adtran encryptors. As a result, application data propagates through the network exclusively in encrypted form, while quantum key generation remains entirely handled by the QKD infrastructure.

The video stream generated by the surveillance system located at the customer office is encrypted employing the secure keys and sent through the OTN over DWDM network towards the FiberCop Central Office. Once the encrypted traffic reaches the FiberCop TEC site, it is decrypted and processed by a cloud-native micro-services application. In the presented use case, the application performs real-time video analytics using an AI model capable of extracting information such as people counting. The results are visualized through a dedicated dashboard, which provides both the processed video stream and the associated analytics, further reinforcing the integration between secure communication and advanced edge applications.

A key enabling component of this architecture is the software-defined QKD Node Agent, a software component implementing the ETSI GS QKD 015 [11] control interface for QKD nodes where QKD devices and Key Management System (KMS) are located. The agent provides a vendor-agnostic abstraction layer, enabling integration of heterogeneous QKD devices within a unified SDN-controlled environment. This is particularly relevant in the proposed demo, where QKD devices, encryption systems, and Edge Cloud resources are orchestrated coherently within a single service framework. The architecture is structured in three layers: a northbound interface exposing RESTCONF and NETCONF APIs [12,13], a core orchestrator maintaining the network state and handling control logic, and a southbound driver layer abstracting vendor-specific implementation. This layered design enables flexibility, interoperability, and seamless integration with SDN controllers, supporting centralized monitoring of link conditions and key performance metrics. The agent also implements an event-driven notification mechanism based on publish-subscribe principles. This enables real-time propagation of events such as QBER threshold violations, link failures, or performance degradation, allowing the SDN controller to react dynamically without relying on continuous polling. The agent does not handle cryptographic material directly, delegating key distribution to ETSI GS QKD 014 [14] implementation, thus improving security and modularity. Additionally, the SDN is able to support both real devices and emulated environments, essential capability during experimental validation within the FiberCop lab environment.

The Edge Cloud infrastructure consists of heterogeneous environments, including OpenStack for control functions and a cloud-native platform hosting the application layer, where services such as video analytics and Internet of Things (IoT) data processing can be executed.

The demonstration starts with an overview of the system architecture and the services provided. The video surveillance application running on the Edge Cloud is shown on a dedicated dashboard providing real-time visualization of both application-level outputs and network-level metrics, including SKR, QBER, and link status. The TeraFlowSDN dashboard enables the observation of the QKD network topology.

Attendees can actively interact with the demo by triggering different network conditions and fault scenarios. First, the optical link used for quantum key generation can be interrupted remotely by a Variable Optical Attenuator (VOA), allowing both to shutter and to attenuate the signal. In this case, the system detects the key generation interruption and raises an alarm, while the service remains operational thanks to the availability of previously generated keys stored in the key database, which continues to feed the encryptors. Subsequently, further actions can be performed, such as interrupting the connection between the encryptor and the QKD system or emptying the key database. In these conditions, the system continues to operate by falling back to classical key alignment mechanism. Throughout these steps, the monitoring framework dynamically reflects system behavior, allowing users to observe how service continuity and security are maintained under different conditions.

## Conclusions

Overall, the demonstration validates complete end-to-end architecture and proves the feasibility of integrating QKD into real network environments, showcasing a full chain from quantum key generation to application-level processing. It highlights how QKD, SDN, and TEC technologies can jointly support secure, low-latency, and resilient services for industrial and mission-critical scenarios. While the validation focuses on a video surveillance use case, the proposed solution is applicable to a wide range of scenarios, including smart manufacturing, critical infrastructure monitoring, robotics, and remote control systems. Through this activity, FiberCop highlights the role of its Innovation Hangar labs as a testing ground for next-generation telecom infrastructures, demonstrating how the integration of QKD, SDN and TEC represents a concrete step toward scalable, resilient, and deployable quantum-safe services.

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[\(QKD\); Protocol and data format of REST-based key delivery API](#)

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